

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406830
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lee; Sun Ryum et al.

Apparatus for treating substrate and method for treating substrate

Abstract

Disclosed is a microwave application unit for applying a microwave to generate plasma. The microwave application unit includes an antenna plate disposed on the support unit and having a plurality of slots; a power supply configured to apply a microwave to the antenna plate; a dielectric plate disposed above the antenna plate to face the antenna plate; an upper plate disposed above the dielectric plate; and a transmissive plate provided below the antenna plate and configured to transmit the microwave to the processing space, wherein an adjustment groove configured to adjust an electric field is formed on an lower surface of the upper plate.

Inventors: Lee; Sun Ryum (Chungcheongnam-do, KR), Chang; Loung Sue (Chungcheongnam-do, KR), Yang; Jung Yoon (Chungcheongnam-do, KR), Lim; In Seong (Gyeonggi-do, KR), Jeon; Seung Hoon (Gyeonggi-do, KR)

Applicant: SEMES CO., LTD. (Chungcheongnam-do, KR)

Family ID: 81994186

Assignee: SEMES CO., LTD. (Chungcheongnam-Do, KR)

Appl. No.: 17/547303

Filed: December 10, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220199364 A1	Jun. 23, 2022

Foreign Application Priority Data

KR	10-2020-0180320	Dec. 21, 2020
----	-----------------	---------------

Publication Classification

Int. Cl.: H01J37/32 (20060101)

U.S. Cl.:

CPC H01J37/3222 (20130101); H01J37/3244 (20130101);

Field of Classification Search

CPC: H01J (37/3222); H01J (37/3244)

USPC: 118/723AN

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2008/0142159	12/2007	Sasaki	257/E21.285	H01J 37/32495
2009/0029564	12/2008	Yamashita	N/A	N/A
2009/0074632	12/2008	Ishibashi	422/186.04	H01J 37/32192
2015/0232993	12/2014	Iwao	118/723AN	H01J 37/32192
2019/0180989	12/2018	Nogami	N/A	H01J 37/32807

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101002509	12/2006	CN	N/A
101133688	12/2007	CN	N/A
101599408	12/2008	CN	N/A
101803472	12/2009	CN	N/A
2010232493	12/2009	JP	N/A
2016225047	12/2015	JP	N/A
10-0485235	12/2004	KR	N/A
10-0824813	12/2007	KR	N/A
10-2011-0128888	12/2010	KR	N/A
10-2013-0035838	12/2012	KR	N/A

OTHER PUBLICATIONS

Office Action dated Feb. 9, 2023 issued by the Korean Patent Office in corresponding KR Patent Application No. 10-2020-0180320, pp. 1-5. cited by applicant

Chinese Office Action issued by the China National Intellectual Property Administration on Mar. 31, 2025 in corresponding CN Patent Application No. 202111572170.3, with English translation. cited by applicant

Primary Examiner: Bennett; Charlee J. C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to and the benefit of the Korean Patent Application No. 10-2020-0180320 filed in the Korean Intellectual Property Office on Dec. 21, 2020, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to an apparatus and a method for treating a substrate, and more particularly, to an apparatus and a method for treating a substrate using plasma.

BACKGROUND ART

(3) Plasma is generated by a very high temperature, a strong electric field, or RF electromagnetic fields, and means an ionized gas condition consisting of ions, electrons, radicals and the like. In a semiconductor device manufacturing process, various processes are performed using plasma. FIG. 1 is a cross-sectional view illustrating a general substrate processing apparatus for processing a substrate using a microwave. Referring to FIG. 1, a substrate W is supported to a support unit 20 disposed in a processing space 12, and the substrate is processed by generating plasma from processing gas by using a microwave in the processing space 12. An antenna plate 40 having a slot 42 is provided in an upper region of the substrate W. A dielectric plate 50 is disposed on the antenna plate 40, and a transmissive plate 60 is disposed below the antenna plate 40. When the microwave is applied to the antenna plate 40, the microwave is propagated along a radial direction of the antenna plate 40 and then transmitted to the processing space 12 through the slot 42 and the transmissive plate 60.

(4) When using a substrate processing apparatus 1 having the structure of FIG. 1, an electric field is concentrated in a region facing an edge region of the antenna plate 40 in the processing space 12, so that the plasma density is relatively higher than that of other regions. Further, in the antenna plate 40 of the processing space 12, the electric field is concentrated in a region facing the region in which the slot 42 is formed, so that plasma density is relatively higher than other regions. In the processing space 12, the non-uniformity of the plasma density deteriorates the process uniformity when the substrate is processed with the plasma. For example, as illustrated in FIG. 2, in the processing space 12, when an edge region A is provided to have higher plasma density than a central region B, an edge region W1 of the substrate has a higher etching rate than a central region W2 of the substrate when performing the etching process. Further, when performing an annealing process for the substrate using the plasma, the annealing degree between the edge region W1 and the central region W2 of the substrate is different from each other.

(5) In order to solve the above problems, the plasma density in the processing space is adjusted by changing a shape of the slot formed in the antenna plate, a position of the slot, and a distribution of the slots, but there is a structural limitation in the change of the slot. Further, the plasma density is adjusted by variously changing the shape of the transmissive plate disposed below the antenna plate, but there is a problem such that particles occur according to the shape of the transmissive plate.

SUMMARY OF THE INVENTION

(6) An object of the present invention is to provide an apparatus for processing a substrate and a method for processing a substrate capable of improving the substrate processing efficiency.

(7) Further, an object of the present invention is to provide an apparatus for processing a substrate and a method for processing a substrate capable of improving the uniformity of plasma density in the entire region of a processing space provided by the substrate.

(8) Further, an object of the present invention is to provide an apparatus for processing a substrate and a method for processing a substrate to which a new factor for adjusting the plasma density is applied in a processing space when processing the substrate using plasma.

(9) Other objects of the present invention are not limited to the objects described above, and other objects, which are not mentioned above, will be apparent to those skilled in the art from the following description.

(10) An exemplary embodiment of the present invention provides an apparatus for treating a substrate. The apparatus for treating the substrate may include a process chamber configured to have a processing space therein; a support unit configured to support the substrate in the process chamber; a gas supply unit configured to supply processing gas to the processing space; and a microwave application unit configured to generate plasma from the processing gas supplied to the processing space. The microwave application unit may include an antenna plate disposed on the support unit and having a plurality of slots; a power supply configured to apply a microwave to the antenna plate; a dielectric plate disposed above the antenna plate to face the antenna plate; an upper plate disposed above the dielectric plate; and a transmissive plate provided below the antenna plate and configured to transmit the microwave to the processing space. An adjustment groove configured to adjust an electric field may be formed on a lower surface of the upper plate.

(11) According to the exemplary embodiment, the adjustment groove may be provided in a region facing the slot formed in the antenna plate.

(12) According to the exemplary embodiment, the adjustment groove may be provided in a ring shape surrounded a center of the antenna plate when viewed from the top.

(13) According to the exemplary embodiment, the adjustment groove may be provided in a region facing an edge region of the antenna plate.

(14) According to the exemplary embodiment, the slots may be arranged to form one ring or a plurality rings to surround the center of the antenna plate, and the adjustment grooves may be provided in regions overlapping with the slots when viewed from the top, respectively.

(15) According to the exemplary embodiment, the slots may be arranged to form one ring or a plurality rings to surround the center of the antenna plate, and the adjustment grooves may be provided so that depths thereof are different from each other according to a distance from the center of the antenna plate.

(16) According to the exemplary embodiment, the adjustment grooves may be provided so that the depths thereof are increased as being far away from the center of the antenna plate.

(17) According to the exemplary embodiment, the upper plate may include a base plate having a through hole; and an adjustment plate disposed in the through hole and provided to be movable in a vertical direction to the base plate, wherein a lower region of the adjustment plate among the through hole is provided as the adjustment groove, and the depth of the adjustment groove is provided to be changeable along the movement in the vertical direction of the adjustment plate.

(18) According to the exemplary embodiment, the upper plate may further include a driver configured to drive the adjustment plate in a vertical direction.

(19) According to the exemplary embodiment, the adjustment plate may be screwed with the base plate.

(20) According to the exemplary embodiment, the upper plate and the dielectric plate may be disposed in contact with each other.

(21) An exemplary embodiment of the present invention provides an apparatus for treating a substrate. The apparatus for treating the substrate includes a process chamber configured to have a processing space therein; a support unit configured to support the substrate in the process chamber; an antenna plate disposed above the support unit and having a plurality of slots penetrated in a vertical direction; a power supply configured to apply a microwave to the antenna plate; a dielectric plate disposed above the antenna plate to face the antenna plate; an upper plate disposed above the dielectric plate and including a cooling member to cool the dielectric plate; a transmissive plate

provided below the antenna plate and configured to transmit the microwave to the processing space; and a gas supply unit configured to supply processing gas to the processing space, wherein a ring-shaped adjustment groove configured to adjust an electric field is formed on an lower surface of the upper plate.

(22) According to the exemplary embodiment, the antenna plate may include a first region and a second region which are provided with the plurality of slots, respectively, and have ring shapes which are concentric with the center of the antenna plate, the first region may be a region which is farther away from the center of the antenna plate than the second region, and the ring-shaped adjustment groove may include a first groove provided in a region facing the first region.

(23) According to the exemplary embodiment, the ring-shaped adjustment groove may further include a second groove provided in a region facing the second region.

(24) According to the exemplary embodiment, a depth of the first groove may be provided larger than a depth of the second groove.

(25) According to the exemplary embodiment, the adjustment groove may be provided in a region facing the slot.

(26) According to the exemplary embodiment, the upper plate may include a base plate provided with a through hole; and an adjustment plate disposed in the through hole and provided to be movable in a vertical direction to the base plate, wherein the depth of the adjustment groove may be provided to be changeable along the movement in the vertical direction of the adjustment plate.

(27) An exemplary embodiment of the present invention provides a method for treating a substrate. The method for processing the substrate includes disposing an upper plate, a dielectric plate, an antenna plate formed with a plurality of slots, and a transmissive plate sequentially in a direction from the top to the bottom, transmitting a microwave applied to the antenna plate to a processing space below the transmissive plate through the transmissive plate, and generating plasma from processing gas supplied to the processing space to treat the substrate in the processing space, wherein adjustment grooves are formed on the lower surface of the upper plate to adjust an electric field provided to the processing space by the microwave.

(28) According to the exemplary embodiment, the processing gas may be gas containing hydrogen, and the treating of the substrate may include an annealing process.

(29) According to the exemplary embodiment, the adjustment groove may be formed in the region facing the slot to reduce the electric field below the slot.

(30) According to the present invention, it is possible to improve the substrate processing efficiency when processing the substrate using the plasma.

(31) According to the present invention, it is possible to improve the uniformity of plasma density in the entire region of the processing space disposed in the substrate.

(32) According to the present invention, it is possible to precisely adjust the plasma density by providing an adjustment groove below an upper plate as a factor for adjusting the plasma density.

(33) The effect of the present invention is not limited to the foregoing effects, and non-mentioned effects will be clearly understood by those skilled in the art from the present specification and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view schematically illustrating a general substrate processing apparatus.
- (2) FIG. 2 is a diagram for describing plasma density for each region in a processing space in the apparatus of FIG. 1.
- (3) FIG. 3 is a cross-sectional view schematically illustrating a substrate processing apparatus

according to an exemplary embodiment of the present invention.

(4) FIG. 4 is a plan view illustrating an example of slots formed in an antenna plate of FIG. 3.

(5) FIG. 5 is a perspective view illustrating an example of an upper plate of FIG. 3.

(6) FIGS. 6 and 7 are a cross-sectional view and a plan view of a microwave application unit illustrating formation positions of adjustment grooves according to an exemplary embodiment of the present invention, respectively.

(7) FIGS. 8 and 9 are a cross-sectional view and a plan view of a microwave application unit illustrating formation positions of adjustment grooves according to another exemplary embodiment of the present invention, respectively.

(8) FIG. 10 is a cross-sectional view of a microwave application unit illustrating formation positions of adjustment grooves according to another exemplary embodiment of the present invention.

(9) FIGS. 11 and 12 are a cross-sectional view and a plan view of a microwave application unit illustrating positions of adjustment grooves according to yet another exemplary embodiment of the present invention, respectively.

(10) FIGS. 13 and 14 are perspective views illustrating an exemplary embodiment of an upper plate provided so that a depth of the adjustment groove is adjustable.

(11) FIGS. 15 and 16 are perspective views illustrating an exemplary embodiment of the upper plate provided so that the depth of the adjustment groove is adjustable.

(12) FIGS. 17 and 18 are diagrams illustrating other modifications of the upper plate, respectively.

DETAILED DESCRIPTION

(13) Hereinafter, an exemplary embodiment of the present invention will be described in more detail with reference to the accompanying drawings. The exemplary embodiment of the present invention can be modified in various forms, and it should not be construed that the scope of the present invention is limited to exemplary embodiments described below. The exemplary embodiments will be provided to more completely describe the present invention to those skilled in the art. Therefore, shapes, and the like of components in the drawings will be exaggerated to emphasize a more clear description.

(14) An apparatus of an exemplary embodiment may be used to perform an annealing process using hydrogen plasma with respect to a circular semiconductor substrate. However, the technical idea of the present invention is not limited thereto, and may be applied to various types of processes for processing the substrate using plasma. For example, the technical idea of the present invention can be applied to an etching process of removing a thin film on a substrate W using plasma or a deposition process of depositing the thin film on the substrate.

(15) Hereinafter, an exemplary embodiment of the present invention will be described with reference to FIGS. 3 to 18.

(16) FIG. 3 is a cross-sectional view schematically illustrating a substrate processing apparatus according to an exemplary embodiment of the present invention. Referring to FIG. 3, a substrate processing apparatus 2 has a process chamber 100, a support unit 140, a microwave application unit 300, and a gas supply unit 400.

(17) The process chamber 100 has a processing space 120 therein. On a side wall of the process chamber 100, a carrying port 162 through which a substrate W is carried is formed. The carrying port 162 may be opened and closed by a door 164.

(18) The support unit 140 may support the substrate W in the processing space 120. According to the exemplary embodiment, the support unit 140 may support the substrate W by an electrostatic force. Optionally, the support unit 140 may support the substrate W by mechanical clamping. Optionally, without a means for fixing the substrate W, the substrate W may be placed on the support unit 140.

(19) The support unit 140 may be provided with a heating member 220 for heating the substrate W. According to the exemplary embodiment, the heating member 220 may be provided as a hot wire

disposed inside the support unit **140**.

(20) The gas supply unit **400** supplies the processing gas to the processing space **120**. The processing gas may include hydrogen. The gas supply unit **400** has a gas supply source **420** and a gas supply line **440**. The gas supply source **420** may be coupled to the side wall of the process chamber **100** by the gas supply line **440**. According to the exemplary embodiment, the side wall of the process chamber **100** is provided with a ring-shaped buffer space **460**, and the gas supply line **440** supplies processing gas to the buffer space **460**. The side wall of the process chamber **100** is formed with an ejection line **480** which is extended from the buffer space **460** to eject the gas to the processing space **120**. A plurality of ejection lines **480** may be provided along the circumference of the process chamber **100**.

(21) An exhaust line **140** is connected to a lower wall of the process chamber **100**. A pump (not illustrated) is connected to the exhaust line **140** to adjust the pressure in the processing space **120** to process pressure. A ring-shaped exhaust baffle **180** is provided on the side of the support unit **140** in the processing space **120**. An inner surface of the exhaust baffle **180** is in contact with the support unit **140**, and an outer surface of the exhaust baffle may be in contact with the side wall of the process chamber **100**. The gas is uniformly exhausted to the lower space from the upper space of the exhaust baffle **180** by the exhaust baffle **180**.

(22) The microwave application unit **300** has a transmissive plate **320**, an antenna plate **340**, a dielectric plate **360**, an upper plate **380**, and a power supply **500**.

(23) The transmissive plate **320** is provided with a quartz material. Optionally, the transmissive plate **320** may be provided with a dielectric material such as aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), sapphire, or silicon nitride (SiN). The transmissive plate **320** functions as an upper wall of the processing space **120** and transmits the microwave to the processing space **120**. According to the exemplary embodiment, the upper surface of the transmissive plate **320** has a flat shape. In addition, a central region of the lower surface of the transmissive plate **320** has a flat shape, and a protrusion **322** protruding downward may be provided in an edge region of the lower surface of the transmissive plate **320**. Optionally, a protrusions or a groove may be formed in the central region of the lower surface of the transmissive plate **320**.

(24) The antenna plate **340** is provided in a disk shape and is disposed on the transmissive plate **320**. The antenna plate **340** is positioned to be in contact with the transmissive plate **320**. Optionally, the antenna plate **340** may be positioned so as to be spaced apart from the transmissive plate **320** at a predetermined distance. The antenna plate **340** may be provided with a metal material. According to the exemplary embodiment, the antenna plate **340** may be provided with a copper or aluminum material. Gold or silver may be plated on the surface of the antenna plate **340**. A plurality of slots **342** for radiating the microwave are formed in the antenna plate **340**. The slots **342** are formed to penetrate from the upper surface to the lower surface of the antenna plate **340**.

(25) FIG. 4 is a plan view illustrating an example of slots formed in the antenna plate.

(26) Referring to FIG. 4, the slots **342** are arranged to form a plurality of rings in the antenna plate **340**. The antenna plate **340** has a first region **344a**, a second region **344b**, and a third region **344c**, which are formed with the slots **342** and have ring shapes. The first region **344a** is positioned farther away from the center of the antenna plate **340** than the second region **344b**. In addition, the second region **344b** is positioned farther away from the center of the antenna plate **340** than the third region **344c**. The first region **344a**, the second region **344b**, and the third region **344c** are provided in a circular ring shape that covers the center of the antenna plate **340**. The first region **344a**, the second region **344b**, and the third region **344b** are concentrically provided to each other. The first region **344a** is provided to cover the second region **344b**, and the second region **344b** is provided to cover the third region **344c**. The first region **344a** and the second region **344b** are spaced apart from each other, and the second region **344b** and the third region **344c** are spaced apart from each other.

(27) Hereinafter, the slots **342** formed in the first region **344a** are referred to as first slots **342**, and

the slots **342** formed in the second region **344b** are referred to as second slots **342**, and the slots **342** formed in the third region **344c** are referred to as third slots **342**.

(28) The first slots **342** are formed in the same shape as each other. According to the exemplary embodiment, the first slot **342** is provided in a '+' shape in which two long '-' shaped slits cross each other. The second slots **342** are formed in the same shape as each other. According to the exemplary embodiment, the second slot **342** is provided in a '+' shape in which two long '-' shaped slits cross each other in the same manner as the first slot **342**. The third slots **342** are formed in the same shape as each other. According to the exemplary embodiment, the third slot **342** has a shape in which four short '-' shaped slits are spaced apart from each other at intervals of 90°.

(29) The shapes of the first slots **342**, the second slots **342**, and the third slots **342** are not limited thereto, and may be variously changed. Further, the number of regions with the slots **342** is not limited thereto and may be variously changed.

(30) The dielectric plate **360** is disposed on the antenna plate **340**. The wavelength of the microwave is changed by the dielectric plate **360**. The dielectric plate **360** is provided with a quartz material. Optionally, the dielectric plate **360** may be provided with a dielectric material such as aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), sapphire, or silicon nitride (SiN). The dielectric plate **360** may be provided with the same material as the transmissive plate **320**.

Optionally, the dielectric plate **360** may be provided with a different material from the transmissive plate **320**. The dielectric plate **360** may be provided in a disk shape. The upper surface and the lower surface of the dielectric plate **360** may be flatly provided, respectively. The dielectric plate **360** may be disposed to be in contact with the antenna plate **340**. Optionally, the dielectric plate **360** may be disposed to be spaced apart from the antenna plate **340** at a predetermined distance.

(31) An upper plate **380** is disposed on the dielectric plate **360**. The upper plate **380** is provided with a metallic material. According to the exemplary embodiment, the upper plate **380** may be provided with an aluminum material. According to the exemplary embodiment, the upper plate **380** may be provided with a cooling plate for cooling the dielectric plate **360**, the antenna plate **340**, and the transmissive plate **320**. A cooling flow channel **381** is formed in the upper plate **380**, and cooling water may flow through the cooling flow channel **381**.

(32) The power supply **500** generates a microwave. The microwave generated in the power supply **500** has a frequency of 23 GHz to 26 GHz. The microwave is transmitted through a waveguide **520**. The waveguide **520** may be provided in a pipe shape with a polygonal cross section. The inner surface of the waveguide **520** is provided as a conductor. For example, the inner surface of the waveguide **520** may be provided with gold or silver.

(33) A coaxial converter **540** is disposed inside the waveguide **520**. The coaxial converter **540** is located on the opposite side of the power supply **500**. One end of the coaxial converter **540** is fixed to the inner surface of the waveguide **520**. The coaxial converter **540** may be provided in a cone shape in which a lower cross-sectional area is smaller than an upper cross-sectional area. The microwave transmitted through the inner space of the waveguide **520** is mode-converted in the coaxial converter **540** to be propagated in a lower direction.

(34) The mode-converted microwave is transferred to an antenna through an outer conductor **560** and an inner conductor **580**. The outer conductor **560** is located between the waveguide **520** and the upper plate **380**. According to the exemplary embodiment, an upper end of the outer conductor **560** is in contact with the waveguide **520**, and a lower end of the outer conductor **560** is in contact with the upper plate **380**. A space connected with an inner space of the waveguide **520** is formed inside the outer conductor **560**. The inner conductor **580** is positioned inside the outer conductor **560**. The inner conductor **580** is provided in a rod shape, and a longitudinal direction is disposed in a vertical direction. The outer surface of the inner conductor **580** is spaced apart from the inner surface of the outer conductor **560**.

(35) An upper end of the inner conductor **580** is fixed to a lower end of the coaxial converter **540**. The inner conductor **580** extends in a lower direction. The lower end of the inner conductor **580** is

fixedly coupled to the center of the antenna plate **340**. The inner conductor **580** is disposed perpendicular to the upper surface of the antenna plate **340**. The inner conductor **580** and the outer conductor **560** are coaxially provided.

(36) The microwave propagated to the antenna plate **340** in the vertical direction is propagated along a radial direction of the dielectric plate **360**, and then transmitted to the processing space **120** through the slots **342** formed in the antenna plate **340** and the transmissive plate **320**.

(37) Referring back to FIG. **4**, an adjustment groove **382** is formed on the upper plate **380**. The adjustment groove **382** adjusts the electric field provided to the processing space **120**. The adjustment groove **382** is formed on the lower surface of the upper plate **380**.

(38) FIG. **5** is a perspective view illustrating an example of the upper plate of FIG. **3**. Referring to FIG. **5**, an adjustment groove **382** may be provided in a ring shape. The adjustment groove **382** is formed at a predetermined depth from the bottom to the top of the upper plate **380**. The adjustment groove **382** may be formed in the same width along a circumferential direction thereof. In addition, the adjustment groove **382** may be formed in the same width along the circumferential direction thereof.

(39) The microwave applied to the antenna plate **340** is propagated outwardly along a radial direction thereof. In the region where the adjustment groove **382** is provided, a part of the microwave is induced to the adjustment groove **382**. As a result, when viewed from the top, an electric field is weakened in a region overlapping with the region in which the adjustment groove **382** is provided in the processing space **120**.

(40) In general, when using the substrate processing apparatus having the general structure as illustrated in FIG. **1**, an electric field is concentrated in a region facing the edge region of the antenna plate in the processing space as the compared with other regions. According to the exemplary embodiment of the present invention, as illustrated in FIG. **3**, the adjustment groove **382** may be formed in the region facing the edge region of the antenna plate **340** of the lower surface of the upper plate **380**.

(41) Further, since the microwave propagated along the antenna plate **340** is transmitted in the lower direction through the slots **342** and the transmissive plate **320**, the electric field is relatively concentrated in the region facing the region formed with the slots **342** in the processing space **120** as compared with other regions. Accordingly, according to the exemplary embodiment, the adjustment groove **382** may be formed in a region facing the region formed with the slots **342** in the lower surface of the upper plate **380**.

(42) The depth and the width of the adjustment groove **382** affect the size of the electric field and the width of the region where the electric field is adjusted. Therefore, the depth and the width of the adjustment groove **382** may be selected in consideration of the size of the electric field and the width of the region where the electric field is concentrated.

(43) FIGS. **6** and **7** are a cross-sectional view and a plan view of a microwave application unit illustrating formation positions of adjustment grooves according to an exemplary embodiment of the present invention, respectively. In FIG. **7**, a region indicated by the shade is a region where the adjustment groove is formed in the upper plate.

(44) Referring to FIGS. **6** and **7**, when the slots **342** are formed in the first region **344a**, the second region **344b**, and the third region **344c** of the antenna plate **340**, an adjustment groove **1382** is formed in a region facing the first region **344a** provided on the outermost side of these regions. The width of the adjustment groove **1382** may be provided substantially the same as the width of the first region **344a**. If the adjustment groove **1382** is not provided, in FIG. **6**, the electric field is most concentrated in a region facing the region where the adjustment groove **1382** is provided in the processing space **12** as a region where the slots **342** are formed while corresponding to the edge region of the antenna plate **340** among regions of the processing space **120** to increase the plasma density. When the adjustment groove **1382** is provided to the region facing the region, it is possible to prevent the electric field from being concentrated in the region. As a result, the plasma

uniformity may be improved in the entire region of the processing space **120**. The formation position of the adjustment groove **1382** in the microwave application unit **1300** described above may be provided differently from those described above depending on the actual density of the plasma for each region in the processing space **120**.

(45) FIGS. **8** and **9** are a cross-sectional view and a plan view of a microwave application unit illustrating formation positions of adjustment grooves according to another exemplary embodiment of the present invention, respectively. In FIG. **8**, a region indicated by the shade is a region where the adjustment groove is formed in the upper plate.

(46) Referring to FIGS. **8** and **9**, an adjustment groove **2382** is formed at a position corresponding to each of regions in which the slots **342** are formed. For example, the slots **342** are formed in the first region **344a**, the second region **344b**, and the third region **344c** of the antenna plate **340**, and the adjustment groove **2382** includes a first groove **2382a**, a second groove **2382b**, and a third groove **2382c**. The first groove **2382a** is formed in a region facing the first region **344a**, the second groove **2382b** is formed in a region facing the second region **344b**, and the third groove **2382c** is formed in a region facing the third region **344c**. According to the exemplary embodiment, the first region **344a**, the second region **344b**, and the third region **344c** may be all provided in the same width as each other, and the first groove **2382a**, the second groove **2382b**, and the third groove **2382c** may be all provided with the same width as each other. Further, both the first groove **2382a**, the second groove **2382b**, and the third groove **2382c**, and the third groove **2382c** may be all provided at the same depth as each other. In the case of using a microwave application unit **2300** of FIGS. **8** and **9**, it is possible to prevent the electric field from being concentrated in the entire region facing the region where the slots **342** are formed in the antenna plate **340** of the processing space **120**.

(47) FIG. **10** is a cross-sectional view of a microwave application unit illustrating formation positions of adjustment grooves according to another exemplary embodiment of the present invention.

(48) A plurality of adjustment grooves **3382** are formed on an upper plate **3380**. Each adjustment groove **3382** is formed in a ring shape. The adjustment grooves **3382** are provided at different depths from each other. According to the exemplary embodiment, the farther the distance from the center of the antenna plate **340**, the larger the depth of the adjustment groove **3382** may be provided. For example, as illustrated in FIG. **10**, a first groove **3382a** is formed on the upper plate **3380** at a position facing the first region **344a**, a second groove **3382b** is formed on the upper plate **3380** at a position facing the second region **344b**, and a third groove **3382c** is formed on the upper plate **3380** at a position facing the third region **344c**. A depth $h1$ of the first groove **3382a** may be provided larger than a depth $h2$ of the second groove **3382b**, and the depth $H2$ of the second groove **3382b** may be provided larger than a depth $h3$ of the third groove **3382c**. In general, since the electric field is further gradually concentrated toward the edge region of the antenna plate **340**, when using a microwave application unit **3300** in FIG. **10**, the electric field may be provided in a more uniform size in the processing space **120**. The size relationship in the depths between the first groove **3382a**, the second groove **3382b**, and the third groove **3382c** may be provided differently above depending on the actual density of the plasma for each region in the process space **120**.

(49) FIGS. **11** and **12** are a cross-sectional view and a plan view of a microwave application unit illustrating positions of adjustment grooves according to yet another exemplary embodiment of the present invention, respectively. In FIG. **12**, a region indicated by the shade is a region where the adjustment groove is formed in the upper plate.

(50) A plurality of adjustment grooves **4382** are formed on an upper plate **4380** in a microwave application unit **4300**. Each adjustment groove **4382** is formed in a ring shape. The adjustment grooves **4382** are provided at different depths from each other. According to the exemplary embodiment, the farther the distance from the center of the antenna plate **340**, the larger the width of the adjustment groove **4382** may be provided. For example, as illustrated in FIG. **10**, a first

groove **4382a** is formed on the upper plate **4380** at a position facing the first region **344a**, a second groove **4382b** is formed on the upper plate **4380** at a position facing the second region **344b**, and a third groove **4382c** is formed on the upper plate **4380** at a position facing the third region **344c**. A width **W1** of the first groove **4382a** may be provided larger than a width **W2** of the second groove **4382b**, and the width **W2** of the second groove **4382b** may be provided larger than a width **W3** of the third groove **4382c**. The size relationship in the widths between the first groove **4382a**, the second groove **4382b**, and the third groove **4382c** may be provided differently above depending on the actual density of the plasma for each region in the process space **120**.

(51) According to yet another exemplary embodiment of the present invention, the adjustment groove on the upper plate may be provided so that the depth thereof is adjustable. FIG. **13** is a perspective view illustrating an exemplary embodiment of an upper plate provided so that the depth of an adjustment groove is adjustable.

(52) Referring to FIG. **13**, an upper plate **5380** has a base plate **5384** and an adjustment plate **5386**. The base plate **5384** is formed with a through hole penetrated in a vertical direction and the adjustment plate **5386** is inserted into the through hole. According to the exemplary embodiment, the through hole is formed in a ring shape, and the adjustment plate **5386** has a ring shape. In the through hole, the lower region of the adjustment plate **5386** functions as the above-described adjustment groove **5382**. The adjustment plate **5386** is provided to be movable in a vertical direction. According to the exemplary embodiment, the adjustment plate **5386** is driven in the vertical direction by a driver **5388**. As illustrated in FIG. **13**, when the adjustment plate **5386** is fully inserted into the through hole, the depth of the adjustment groove **5382** is provided small. As illustrated in FIG. **14**, when a part of the adjustment plate **5386** is moved to protrude to the top of the base plate **5384**, the depth of the adjustment groove **5382** becomes larger. The driver **5388** is provided as a cylinder so that the depth of the adjustment groove **5382** may be changed between a first size and a second size. Optionally, the driver **5388** is provided as a motor so that the depth of the adjustment groove **5382** may be changed in various sizes.

(53) Optionally, the depth adjustment of an adjustment groove **6382** may be made manually by an operator. Threads may be formed on the inner surface of the through hole of the base plate **6384** and the outer surface of the adjustment plate **6386**, respectively. A tool groove **6388** to which a tool is inserted may be formed on the upper surface of the adjustment plate **6386** to rotate the adjustment plate **6386**. As illustrated in FIG. **15**, when the adjustment plate **6386** is fully inserted into the through hole, the depth of the adjustment groove **6382** is provided small. As illustrated in FIG. **16**, when a part of the adjustment plate **6386** is moved to protrude to the top of the base plate **6384** by the rotation of the adjustment plate **6386**, the depth of the adjustment groove **6382** is provided large.

(54) In the above exemplary embodiment, the case where the adjustment groove is formed in a ring shape has been described as an example. However, unlike this, as illustrated in FIG. **17**, an upper plate **7380** is provided with a plurality of adjustment grooves **7382** at the same distance from the center of the antenna plate **340**, and these adjustment grooves **7382** may be arranged to form a ring.

(55) In the above exemplary embodiment, it has been described as an example that the adjustment grooves are arranged based on the distance from the center of the antenna plate. However, unlike this, when the region where the electric field is concentrated in the processing space **120** is a specific region regardless of the distance, in an upper plate **8380**, as illustrated in FIG. **18**, an adjustment groove **8382** may be formed only in a region facing the above specific region.

(56) In the above exemplary embodiment, it has been described that the adjustment grooves are formed in the region facing the slots formed in the antenna plate. However, unlike this, the adjustment grooves may be formed in a region that is not overlapped with the slots formed on the antenna plate when viewed from the top.

(57) The foregoing detailed description illustrates the present invention. Further, the above content shows and describes the exemplary embodiment of the present invention, and the present invention

can be used in various other combinations, modifications, and environments. That is, the foregoing content may be modified or corrected within the scope of the concept of the invention disclosed in the present specification, the scope equivalent to that of the disclosure, and/or the scope of the skill or knowledge in the art. The foregoing exemplary embodiment describes the best state for implementing the technical spirit of the present invention, and various changes required in specific application fields and uses of the present invention are possible. Accordingly, the detailed description of the invention above is not intended to limit the invention to the disclosed exemplary embodiment. Further, the accompanying claims should be construed to include other exemplary embodiments as well.

Claims

1. An apparatus for treating a substrate comprising: a process chamber configured to have a processing space therein; a support configured to support the substrate in the process chamber; a gas supply configured to supply processing gas to the processing space; and a microwave application unit configured to generate plasma from the processing gas supplied to the processing space, wherein the microwave application unit includes: a transmissive plate configured to transmit a microwave to the processing space; an antenna plate disposed on the support unit transmissive plate and having a plurality of slots; a power supply configured to apply the microwave to the antenna plate; a dielectric plate disposed above the antenna plate to face the antenna plate; and an upper plate disposed above the dielectric plate, wherein an adjustment groove is formed on a lower surface of the upper plate and configured to adjust an electric field, wherein the adjustment groove is configured to affect the size and width of the electric field by adjusting a depth and width of the adjustment groove.
2. The apparatus for treating the substrate of claim 1, wherein the adjustment groove is provided in a region facing the plurality of slots formed in the antenna plate.
3. The apparatus for treating the substrate of claim 1, wherein the adjustment groove is provided in a ring shape surrounding a center of the antenna plate when viewed from the top.
4. The apparatus for treating the substrate of claim 1, wherein the adjustment groove is provided in a region facing an edge region of the antenna plate.
5. The apparatus for treating the substrate of claim 1, wherein the plurality of slots are arranged to form one ring or a plurality of rings surrounding the center of the antenna plate, and wherein the adjustment groove is provided in regions overlapping with the plurality of slots when viewed from the top, respectively.
6. The apparatus for treating the substrate of claim 1, wherein the plurality of slots are arranged to form one ring or a plurality of rings to surround the center of the antenna plate, and wherein the adjustment groove is provided so that depths thereof are different from each other according to a distance from the center of the antenna plate.
7. The apparatus for treating the substrate of claim 6, wherein the adjustment grooves are provided so that the depths thereof are increased as being far away from the center of the antenna plate.
8. The apparatus for treating the substrate of claim 1, wherein the upper plate includes a base plate having a through hole; and an adjustment plate disposed in the through hole and provided to be movable in a vertical direction to the base plate, wherein a lower region of the adjustment plate among the through hole is provided as the adjustment groove, and wherein the depth of the adjustment groove is provided to be changeable along the movement in the vertical direction of the adjustment plate.
9. The apparatus for treating the substrate of claim 8, wherein the upper plate further includes a driver configured to drive the adjustment plate in a vertical direction.
10. The apparatus for treating the substrate of claim 8, wherein the adjustment plate is screwed with the base plate.

11. The apparatus for treating the substrate of claim 1, wherein the upper plate and the dielectric plate are disposed in contact with each other.

12. An apparatus for treating a substrate comprising: a process chamber configured to have a processing space therein; a support configured to support the substrate in the process chamber; an antenna plate disposed above the support unit and having a plurality of slots penetrated in a vertical direction; a power supply configured to apply a microwave to the antenna plate; a dielectric plate disposed above the antenna plate to face the antenna plate; an upper plate disposed above the dielectric plate and including a cooling flow channel to cool the dielectric plate; a transmissive plate provided below the antenna plate and configured to transmit the microwave to the processing space; and a gas supply configured to supply processing gas to the processing space, wherein a ring-shaped adjustment groove is formed on a lower surface of the upper and configured to adjust an electric field.

13. The apparatus for treating the substrate of claim 12, wherein the antenna plate includes a first region and a second region which are provided with the plurality of slots, respectively, and have ring shapes which are concentric with the center of the antenna plate, wherein the first region is a region which is farther away from the center of the antenna plate than the second region, and wherein the ring-shaped adjustment groove includes a first groove provided in a region facing the first region.

14. The apparatus for treating the substrate of claim 13, wherein the ring-shaped adjustment groove further includes a second groove provided in a region facing the second region.

15. The apparatus for treating the substrate of claim 14, wherein a depth of the first groove is provided larger than a depth of the second groove.

16. The apparatus for treating the substrate of claim 12, wherein the adjustment groove is provided in a region facing the plurality of slots.

17. The apparatus for the substrate of claim 12, wherein the upper plate includes a base plate provided with a through hole; and an adjustment plate disposed in the through hole and provided to be movable in a vertical direction to the base plate, wherein the depth of the adjustment groove is provided to be changeable along the movement in the vertical direction of the adjustment plate.
